

Bell Without Faster Than Light

Equations of the Time-Phase Clock Interpretation

A Radiologist's Toy Theory

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Abstract

The phase of the Schrödinger wave function is interpreted as an internal clock carried by each particle, oscillating at $\omega = E/\hbar$. Entanglement is synchronized clocks at creation. Measurement is Kuramoto synchronization with a local detector. The wave never collapses. No faster-than-light signal is required. This document collects the key equations developed across the project, from the Bell correlation through Dirac spinors, the Higgs mechanism, gravity, Penrose objective reduction, twistor theory, thermodynamics, Brownian motion, and the Heisenberg uncertainty principle.

Contents

The Time-Phase Wave Function

The Schrödinger equation – which describes orbitals exactly and never needs to collapse:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi \quad (1)$$

The wave function augmented with an explicit internal clock phase φ :

$$\psi_A(\mathbf{x}, t) = \sum_s c_s |s\rangle \cdot e^{-i\omega t + i\varphi_A(t)} \quad (2)$$

where $\omega = E/\hbar$ is the particle's clock frequency and $\varphi_A(t)$ is its internal clock phase, distinct from the dynamical phase $-i\omega t$.

The probability of detecting a property along axis $\hat{n}$ depends on the **relative phase** between particle clock φ_A and detector clock Φ_D :

$$P(+1 | \hat{n}, \varphi_A, \Phi_D) = \cos^2\left(\frac{\varphi_A - \Phi_D - \theta_{\hat{n}}}{2}\right) \quad (3)$$

Entanglement is synchronized clocks at the moment of creation:

$$\varphi_A(0) = \varphi_B(0) = \varphi_0 \quad (\text{shared initial phase, one spacetime point}) \quad (4)$$

Kuramoto Phase Synchronization

The Kuramoto (1975) equation governs clock synchronization and thereby measurement:

$$\frac{d\varphi}{dt} = \omega + K \sin(\Phi_{\text{bulk}} - \varphi) \quad (5)$$

- Free flight: $K = 0 \Rightarrow \varphi(t) = \varphi_0 + \omega t$
- At detector: $K > 0 \Rightarrow \varphi \rightarrow \Phi_{\text{bulk}}$ (synchronization)

Synchronization fixed point: $\varphi^* = \Phi_{\text{bulk}}$. Synchronization timescale: $\tau_{\text{sync}} = 1/K$.

Both detectors are pre-synchronized by thermal equilibration with the macroscopic environment:

$$\Phi_{D_1} = \Phi_{D_2} = \Phi_{\text{bulk}} \quad (\text{established locally, before the experiment}) \quad (6)$$

Bell's Theorem – Where the Factorization Fails

Bell's hidden variable theorem:

$$E(a, b) = \int A(a, \lambda) B(b, \lambda) \rho(\lambda) d\lambda \quad (7)$$

The hidden state in this model is $\lambda = (\varphi_0, \Phi_{\text{bulk}})$. Because Φ_{bulk} is the *same* variable at both detectors, the joint distribution does not factorize:

$$\rho(\varphi_0, \Phi_{\text{bulk}}) \neq \rho(\varphi_0) \cdot \rho(\Phi_{\text{bulk}}) \quad (8)$$

Bell's integral (??) implicitly assumes independent components. This is not a conspiracy: Φ_{bulk} is a single shared variable established subliminally and locally, long before the measurement.

Assumption	Standard form	Status in this model
Locality	A depends only on (a, λ)	Satisfied — each detector is local
Measurement independence	$\rho(\lambda \mid a, b) = \rho(\lambda)$	Fails — $\Phi_{\text{bulk}} \in \lambda$ is shared

Correlation functions

Model	$E(a, b)$	CHSH _{max}
Classical LHV (deterministic)	$-(1 - 2 \Delta /\pi)$	2.000
Phase-clock (Malus law)	$-\frac{1}{2} \cos \Delta$	$\sqrt{2} \approx 1.414$
Standard quantum mechanics	$-\cos \Delta$	$2\sqrt{2} \approx 2.828$

Dirac Spinors as Time-Phase / Property Rotation

A positive-energy Dirac spinor with momentum p along z (natural units):

$$u(p, \uparrow) = N \begin{pmatrix} 1 \\ 0 \\ r \\ 0 \end{pmatrix}, \quad u(p, \downarrow) = N \begin{pmatrix} 0 \\ 1 \\ 0 \\ -r \end{pmatrix}, \quad r = \frac{p}{E + m}, \quad N = \sqrt{\frac{E + m}{2E}} \quad (9)$$

The **mixing angle** between temporal and spatial clock vectors:

$$\tan \frac{\theta_{\text{rel}}}{2} = \frac{|\mathbf{p}|}{E + m} \iff \sin \theta_{\text{rel}} = \frac{v}{c} \quad (10)$$

Limits:

$$\theta_{\text{rel}} \xrightarrow{v/c \rightarrow 0} 0 \quad (\text{NR: Pauli spinor sufficient}), \quad \theta_{\text{rel}} \xrightarrow{v/c \rightarrow 1} \frac{\pi}{2} \quad (\text{massless: permanent orthogonality})$$

Three-term decomposition of the Bell correlation

$$E(a, b) = \underbrace{E_{LL}}_{\text{temporal} \times \text{temporal}} + \underbrace{E_{SS}}_{\text{spatial} \times \text{spatial}} + \underbrace{E_{LS}}_{\text{rotation coupling}} = -\cos(a - b) \quad (11)$$

The NR clock model retains only $E_{LL} \approx -\frac{1}{2} \cos(a - b)$. The Dirac small component ($E_{SS} + E_{LS}$) supplies the missing half, verified numerically to machine precision ($< 10^{-15}$) for all momenta.

Massless Particles – Permanent Orthogonality

For $m = 0$: $\theta_{\text{rel}} = \pi/2$ always. The Dirac equation decouples:

$$i\sigma^\mu \partial_\mu \chi_R = 0 \quad (\text{right-handed Weyl: spatial clock, } h = +\frac{1}{2}) \quad (12)$$

$$i\bar{\sigma}^\mu \partial_\mu \chi_L = 0 \quad (\text{left-handed Weyl: temporal clock, } h = -\frac{1}{2}) \quad (13)$$

The two clocks are permanently orthogonal and never mix. This is why photons have exactly **2** polarization states rather than 4.

Higgs Field as Kuramoto Synchronizer

The Yukawa interaction written in Weyl (clock) form:

$$\frac{d\varphi_L}{dt} = \omega + K \sin(\varphi_R - \varphi_L + \delta_{\text{CP}}), \quad \frac{d\varphi_R}{dt} = \omega + K \sin(\varphi_L - \varphi_R - \delta_{\text{CP}}) \quad (14)$$

where the **Kuramoto coupling equals the fermion mass**:

$$K = y_f \cdot \frac{v}{\sqrt{2}} = m_f \quad (15)$$

y_f is the Yukawa coupling, $v = 246 \text{ GeV}$ is the Higgs vacuum expectation value. Without the Higgs ($K = 0$) all fermions are massless and $\theta_{\text{rel}} = \pi/2$.

Antiparticles as reversed clocks

	Particle $u(p)$	Antiparticle $v(p)$
Phase	$e^{-iEt/\hbar}$	$e^{+iEt/\hbar}$
Clock direction	Forward ($+E$)	Reversed ($-E$)
Large/small components	Temporal dominant	Spatial dominant (swapped)
Sync equilibrium	$\varphi_L - \varphi_R \rightarrow +\delta_{\text{CP}}$	$\varphi_L - \varphi_R \rightarrow -\delta_{\text{CP}}$

Matter-antimatter asymmetry from CP-violating synchronization:

$$\eta = \frac{N_{\text{matter}} - N_{\text{antimatter}}}{N_{\text{total}}} \approx \varepsilon \cdot \sin^2(\delta_{\text{CP}}) \quad (16)$$

Gravity as Bulk Clock Synchronization

Clock rate in a gravitational field

$$\omega(\mathbf{x}) = \omega_\infty \sqrt{1 + \frac{2\Phi(\mathbf{x})}{c^2}} \approx \omega_\infty \left(1 + \frac{\Phi}{c^2}\right), \quad \Phi = -\frac{GM}{r} \quad (17)$$

Poisson equation = Kuramoto field equation

$$\underbrace{\nabla^2 \Phi = 4\pi G \rho}_{\text{Poisson (gravity)}} \longleftrightarrow \underbrace{\nabla^2 \varphi_{\text{clock}} = -\rho K_{\text{grav}}}_{\text{Kuramoto (sync field)}} \quad (18)$$

The gravitational potential *is* the bulk clock synchronization field. The clock-synchronization force recovers Newton exactly:

$$a_{\text{clock}} = \frac{d\omega}{dr} \cdot \frac{c^2}{\omega_0} = \frac{GM}{r^2} \quad \checkmark \quad (19)$$

Penrose objective reduction as gravitational clock decoherence

Phase difference accumulated by superposition $|x_1\rangle + |x_2\rangle$:

$$\delta\varphi(t) = [\omega(x_1) - \omega(x_2)] \cdot t \quad (20)$$

Superposition coherence:

$$\mathcal{C}(t) = \cos\left(\frac{\delta\varphi(t)}{2}\right) \quad (21)$$

Collapse when $\mathcal{C} \rightarrow 0$ (i.e. $\delta\varphi = \pi$):

$$\tau_{\text{collapse}} = \frac{\pi\hbar}{E_G}, \quad E_G = \frac{Gm^2}{\Delta x} \quad (22)$$

Equation (??) is the Penrose formula, here derived from gravitational clock decoherence. No observer is required.

Twistor Theory Connection

Penrose's twistor $Z^\alpha = (\omega^A, \pi_{A'})$ maps to:

$$\omega^A \longleftrightarrow \psi_L \text{ (temporal clock)}, \quad \pi_{A'} \longleftrightarrow \psi_R \text{ (spatial clock)}$$

Null twistor condition (massless):

$$Z \cdot \bar{Z} = \omega^A \bar{\pi}_A + \bar{\omega}^{A'} \pi_{A'} = 0 \quad \Longleftrightarrow \quad \theta_{\text{rel}} = 90^\circ \quad \Longleftrightarrow \quad m = 0 \quad (23)$$

Incidence relation (how spacetime position couples the two clocks):

$$\omega^A = i x^{AA'} \pi_{A'} \quad (\text{massless}) \quad (24)$$

$$\omega^A = i x^{AA'} \pi_{A'} + m \eta^A \quad (\text{massive: Higgs coupling term}) \quad (25)$$

The massive incidence relation is the Dirac equation in Weyl form. Gravity curves twistor space by deforming the incidence relation.

Temperature as Clock Phase Distribution Width

$$P(\varphi) \propto \exp\left(-\frac{\hbar\omega \varphi^2}{2k_B T}\right) \quad (\text{classical limit } k_B T \gg \hbar\omega) \quad (26)$$

Clock phase noise standard deviation:

$$\sigma_\varphi(T) = \sqrt{\frac{k_B T}{\hbar\omega}}, \quad \sigma_\varphi(0) = \frac{1}{\sqrt{2}} \text{ rad} \quad (\text{zero-point floor}) \quad (27)$$

Bose-Einstein occupation number:

$$\langle n(\omega) \rangle = \frac{1}{e^{\hbar\omega/k_B T} - 1} \quad (28)$$

Planck blackbody spectrum from thermal clock distribution:

$$u(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3} \cdot \frac{1}{e^{\hbar\omega/k_B T} - 1} + \underbrace{\frac{\hbar\omega^3}{2\pi^2 c^3}}_{\text{zero-point residual}} \quad (29)$$

Absolute zero is unreachable: the zero-point residual field ($\frac{1}{2}\hbar\omega$ per mode) is the minimum clock noise floor.

Brownian Motion from Clock Desynchronization

At each molecular collision the momentum impulse is proportional to the clock phase difference at contact:

$$\delta p = K \sin(\varphi_i - \varphi_j) \approx K \delta\varphi, \quad \delta\varphi \sim \mathcal{N}(0, \sigma_\varphi^2(T)) \quad (30)$$

Mean squared impulse per collision:

$$\langle (\delta p)^2 \rangle = K^2 \sigma_\varphi^2(T) = K^2 \frac{k_B T}{\hbar\omega} \quad (31)$$

The Langevin equation for a Brownian particle:

$$m\ddot{x} = -\gamma\dot{x} + \xi(t), \quad \langle \xi(t) \xi(t') \rangle = 2\gamma k_B T \delta(t - t') \quad (32)$$

Diffusion coefficient predicted from clock parameters:

$$D_{\text{clock}} = \frac{K^2 k_B T}{\hbar\omega} \cdot \frac{\tau_{\text{coll}}}{2m^2} \propto T \quad (33)$$

Matching Eq. (??) to the Stokes-Einstein relation $D = k_B T / (6\pi\eta r)$ fixes the Kuramoto coupling:

$$K = m \sqrt{\frac{\hbar\omega}{3\pi\eta r \tau_{\text{coll}}}} \quad (34)$$

New testable prediction

Standard Stokes-Einstein has no dependence on molecular vibration frequency ω . The clock model predicts:

$$D \propto \frac{1}{\sqrt{\omega}} \quad \text{for fixed } \eta, r, \tau \quad (35)$$

Measurable by single-particle tracking across solvents of different molecular mass.

Quantum-Classical Transition

Nelson's stochastic mechanics

Nelson (1966) proved: Brownian motion in a background field with diffusion $\nu = \hbar/2m$ satisfies the Schrödinger equation exactly. The time-phase model identifies that background field as the zero-point residual wave field.

Total diffusion (quantum + thermal):

$$\nu_{\text{total}} = \underbrace{\frac{\hbar}{2m}}_{\text{quantum (ZPF)}} + \underbrace{\frac{k_B T \tau}{m}}_{\text{thermal (Brownian)}} \quad (36)$$

The quantum-classical crossover:

$$\frac{k_B T \tau}{m} \gg \frac{\hbar}{2m} \Rightarrow \text{classical Brownian}, \quad \frac{k_B T \tau}{m} \ll \frac{\hbar}{2m} \Rightarrow \text{quantum (Schrödinger)} \quad (37)$$

The Schrödinger's cat boundary coincides with the Penrose collapse time equalling the thermal decoherence time – two criteria derived from the same clock model.

$$\tau_{\text{Penrose}} = \frac{\pi \hbar \Delta x}{G m^2} \iff \tau_{\text{thermal}} = \frac{m}{k_B T \tau_{\text{coll}}} \quad (38)$$

System	$\hbar/2m$ (m ² /s)	$k_B T \tau / m$ at 300 K	Regime
Electron in atom ($\tau \sim 10^{-15}$ s)	3×10^{-5}	10^{-21}	Quantum
Proton ($\tau \sim 10^{-13}$ s)	1.7×10^{-8}	10^{-18}	Quantum
Protein (10^{-22} kg)	5×10^{-13}	10^{-14}	Borderline
1 μm dust grain	10^{-22}	10^{-11}	Classical

The Born Rule from Synchronization Statistics

The Born rule — $P = |\psi|^2$ — is derived from Kuramoto synchronization statistics rather than postulated.

The Kuramoto order parameter

For N coupled oscillators with phases $\varphi_1, \dots, \varphi_N$, the synchronization amplitude is:

$$R \cdot e^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\varphi_j} \quad (39)$$

The probability that a single oscillator locks to the collective is proportional to R^2 — the squared modulus of the complex order parameter.

Application to measurement

A particle in superposition $\psi = \alpha|0\rangle + \beta|1\rangle$ arrives at a detector with bulk phase Φ_{bulk} . The synchronization probability for each component depends on its complex coupling amplitude. Averaging over the uniformly distributed bulk phase eliminates cross terms:

$$\langle e^{i(\varphi_0 - \varphi_1)} \rangle_{\Phi_{\text{bulk}}} = 0 \quad (40)$$

Only the diagonal terms survive:

$$P(|0\rangle) = |\alpha|^2, \quad P(|1\rangle) = |\beta|^2 \quad (41)$$

Why the squared modulus

The wave function is complex because the internal clock is a phase oscillator living on a circle in the complex plane. The probability of synchronization is the squared modulus because:

1. Coupling is linear in the complex amplitude (Kuramoto sine coupling)
2. Probability is quadratic in the coupling (energy transfer $\propto$ amplitude²)
3. The complex phase averages out over the bulk, leaving only |amplitude|²

The Born rule is the statement that quantum probability equals wave intensity — the natural measure for any physical wave coupling to a resonant detector.

Heisenberg Uncertainty from Clock Complementarity

The uncertainty principle is not an axiom in this framework — it is a geometric consequence of the orthogonality between temporal and spatial clocks.

Position-momentum uncertainty

The temporal clock oscillates at frequency $\omega = E/\hbar$; the spatial clock oscillates at wavenumber $k = p/\hbar$. These are coupled by the incidence relation (Section 8):

$$\omega^A = i x^{AA'} \pi_{A'} + m \eta^A \quad (42)$$

A sharp position measurement locks the spatial clock phase φ_R to a definite value at a point x . But the incidence relation couples φ_R to φ_L through the spacetime position — locking one forces the other into a superposition of frequencies. This is the Fourier relationship between position and momentum representations, here derived from clock geometry rather than postulated.

The mixing angle θ_{rel} encodes the tradeoff. When $\theta_{\text{rel}} \rightarrow 0$ (rest frame), the temporal clock dominates and energy is sharp — but position is maximally uncertain. When $\theta_{\text{rel}} \rightarrow \pi/2$ (ultrarelativistic), the spatial clock dominates and momentum is sharp — but energy-time is maximally uncertain.

The minimum uncertainty product follows from the Kuramoto coupling between the two clocks. The synchronization force

$$\frac{d\varphi_L}{dt} = \omega + K \sin(\varphi_R - \varphi_L) \quad (43)$$

ensures that locking φ_L (energy measurement) introduces noise in φ_R (momentum) of order $\delta\varphi_R \geq 1/(2\delta\varphi_L)$, because the sine coupling transfers phase uncertainty bidirectionally. Converting to physical units ($\Delta E = \hbar \Delta\omega$, $\Delta p = \hbar \Delta k$):

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad (44)$$

Energy-time uncertainty

The particle clock runs at $\omega = E/\hbar$. A measurement lasting time Δt samples $\Delta t/(2\pi/\omega)$ oscillation cycles. The frequency resolution of any oscillator observed for time Δt is $\Delta\omega \geq 1/(2\Delta t)$. Multiplying by $\hbar$:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad (45)$$

This is identical to the Fourier bandwidth theorem — but here the “signal” is a physical clock, not an abstract wave.

Zero-point phase noise as the uncertainty floor

From Section 9, the zero-point phase noise at $T = 0$ is $\sigma_\varphi(0) = 1/\sqrt{2}$ rad. This residual prevents perfect phase locking of either clock. The zero-point energy $\frac{1}{2}\hbar\omega$ per mode is the energy cost of this minimum phase uncertainty. The uncertainty principle is thus the statement that the zero-point field cannot be removed — complete synchronization of both clocks simultaneously is forbidden by the geometry of spinor space.

Nyquist–Shannon bandwidth limit

The connection between the zero-point floor and the uncertainty principle can be made sharper through a Nyquist–Shannon argument. A Dirac particle’s internal clock oscillates at the Compton frequency $\omega_C = mc^2/\hbar$. By the Nyquist sampling theorem, this clock has a maximum spatial bandwidth: it can resolve phase structures no finer than the Compton wavelength $\lambda_C = h/(mc)$. This sets an irreducible minimum position uncertainty:

$$\Delta x_{\min} \sim \frac{\hbar}{mc} \quad (46)$$

For a particle with momentum spread Δp , the de Broglie relation gives a spatial frequency $k = p/\hbar$ that the clock must resolve. The Nyquist limit requires the clock bandwidth $mc/\hbar$ to exceed k , yielding:

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad (47)$$

The uncertainty principle is thus a **bandwidth limitation of the particle’s own internal clock**. A heavier particle has a faster clock and therefore finer spatial resolution, but the resolution is always finite. No oscillator can resolve frequencies beyond its own Nyquist limit.

The commutator from clock coupling

The canonical commutation relation

$$[\hat{x}, \hat{p}] = i\hbar \quad (48)$$

emerges from the non-commutativity of sequential clock measurements. Measuring position (locking φ_R , then reading φ_L) and measuring momentum (locking φ_L , then reading φ_R) do not commute because the Kuramoto sine coupling is nonlinear — the order of synchronization matters. The imaginary unit i reflects the $\pi/2$ phase shift between the two clocks at the point of maximum complementarity ($\theta_{\text{rel}} = \pi/2$).

Complete Dictionary

Time-phase model	Standard physics	Geometric (Twistor)
Temporal clock ψ_L	Left Weyl spinor	ω^A
Spatial clock ψ_R	Right Weyl spinor	$\pi_{A'}$
L-R coupling $= m$	Dirac mass term	Non-null: $Z \cdot \bar{Z} \neq 0$
$\theta_{\text{rel}} = 90^\circ$	Massless particle	Null twistor
Higgs synchronizer	Yukawa coupling	Twistor cohomology
Gravity $= \nabla^2 \varphi$	Newtonian potential	Twistor space curvature
Penrose collapse	Wavefunction collapse	$\delta\varphi = \pi$
Temperature	$k_B T$	$\sigma_\varphi^2 = k_B T / \hbar \omega$
Brownian motion	Stokes-Einstein	$D = K^2 \sigma_\varphi^2 \tau / 2m^2$
Zero-point field	$\frac{1}{2} \hbar \omega$ per mode	$\sigma_\varphi(0) = 1/\sqrt{2}$
Antiparticle	$e^{+iEt/\hbar}$	Reversed clock
Matter asymmetry	Baryogenesis	$\eta \approx \varepsilon \sin^2 \delta_{\text{CP}}$
Born rule $P = \psi ^2$	Postulated	Sync probability $= R ^2$
Uncertainty principle	$\Delta x \Delta p \geq \hbar/2$	Clock orthogonality: $\delta\varphi_L \delta\varphi_R \geq \frac{1}{2}$
Measurement	Wavefunction collapse	Kuramoto re-sync to bulk
Entanglement	Non-separable state	Synchronized clocks: $\varphi_A = \varphi_B$
Bell's theorem	No local HV model	Φ_{bulk} shared $\Rightarrow$ factorization fails
Quantum-classical	Decoherence	$k_B T \tau / m \gg \hbar / 2m$
Nelson stochastic QM	$\nu = \hbar / 2m$	Zero-point field diffusion

Generated from conversations between a radiologist and Claude (Anthropic). Mathematical framework developed by AI; physical intuitions by the author. Inspirations: de Broglie (pilot wave), Bohm (holographic order), Penrose (twistors, OR, CCC), Kuramoto (phase synchronization), Nelson (stochastic QM).